

Uniform Population Fairness: Common Kernels and Finite-Sample Utility

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Abstract

Population fairness constrains an unknown distribution, whereas a learning algorithm acts on finitely many observations. We study the resulting distinction between exact equality and positive tolerance in finite statistical experiments. In a two-group binary model with known noiseless labels, we characterize the central improvement over constant prediction attainable by learners that satisfy demographic parity uniformly over the population family. At fixed fairness and utility confidence levels, this improvement has order $\min\{1, \epsilon\sqrt{n}\}$, with explicit finite-sample constants. The upper bound combines the geometry of approximate-fairness strips with a local comparison of experiments; a confidence-calibrated shrinkage rule supplies the lower bound. A polynomial common-kernel characterization explains the zero-tolerance endpoint for arbitrary randomized learners. In a multigroup affine family, the same characterization gives an exact central expected-risk cost determined by label approximation within the known uncertainty kernel. Synthetic experiments examine the utility scale, an exact-fair oracle comparison, fixed-rank variation in label cost, and sensitivity to omitted uncertainty directions. The results isolate a tractable statistical mechanism under prescribed finite-alphabet geometry and known labels.

1 Introduction

Demographic parity requires equal population acceptance rates across protected groups. A learner can enforce equality of empirical acceptance rates, but the population equalities depend on quantities that the training sample only estimates. This distinction becomes especially consequential when equality must hold exactly, with a fixed high probability, at every population in a specified family. At a population where the label itself has identical groupwise frequencies, perfect prediction may be fair. The same rule can be unfair at a nearby population that is statistically difficult to distinguish. Understanding the cost of a uniform fairness requirement therefore requires comparing distributions, rather than examining the feasible set at one distribution alone.

Fair learning already contains substantial theory for this distinction. Woodworth et al. (2017) analyze learning under approximate equalized odds and discuss the statistical obstruction associated with exact constraints. Reductions methods translate empirical constraint control into population guarantees with sampling slack (Agarwal et al., 2018). Finite-sample post-processing methods also permit positive fairness tolerances (Li et al., 2023). These precedents motivate a more specific question: in a finite population family, how does the useful prediction attainable under a uniform demographic-parity requirement change as the tolerance crosses the sampling scale?

We examine this question first in a deliberately explicit experiment. Each of two groups supplies n independent binary observations. The group feature frequencies are $1/2 + u$ and $1/2 + v$, the label is the known function $Y = X$, and the test groups have equal weight. The unknown parameter (u, v) ranges over an open square. The learner may use arbitrary randomization and return group-dependent randomized predictions. Fairness is assessed for the realized prediction rule, conditional on that rule; the confidence probability averages over the training data and the learner’s randomness.

At the central population $u = v = 0$, the constant predictor has risk $1/2$, while the known label rule has risk zero and is exactly fair. Let G_n denote the largest improvement over $1/2$ that some uniformly fair learner achieves at the center with a prescribed utility confidence. For fixed confidence parameters whose failure probabilities sum to less than one, Theorem 4 gives

$$G_n(\epsilon; \delta, \eta) \asymp_{\delta, \eta} \min\{1, \epsilon\sqrt{n}\}.$$

The statement is a finite-model central utility characterization. The fairness requirement is global over the population family, while the utility criterion is a lower quantile at one population. The constants are explicit, and the quantifiers allow a different learner for each sample size and tolerance.

The main obstacle in the converse is to relate usefulness under the central output law to fairness at nearby populations. A useful rule has a nonnegligible slope in at least one group’s feature response. Its approximately fair parameter set consequently occupies only a strip of a small parameter square. The output law at the center must be fixed before integrating over that square; replacing it with a parameter-dependent law would invalidate the geometric argument. A total-variation comparison of the training experiments controls the replacement error. This yields Theorem 3, from which the central envelope follows by a radius choice of order $n^{-1/2}$. A shrinkage construction supplies the matching order while satisfying fairness throughout the family.

Exact fairness has an additional algebraic description. Theorem 1 applies to finite observation experiments with continuous likelihoods and polynomial equality constraints on a compact prediction set. A uniformly exactly fair learner must put the required probability mass on the intersection of all population constraint kernels. The proof integrates polynomial zero sets against the learner’s output measures and then uses continuity to move from almost every parameter to every parameter. In the binary example, this intersection contains only constant predictors.

The multigroup construction makes that dependence explicit. Group distributions vary independently within a known affine uncertainty span. Theorem 2 identifies a generally nonconstant common kernel and computes the optimal central expected risk among uniformly exactly fair learners. The cost is a label-approximation problem within that kernel. A four-cell family keeps the kernel dimension fixed while varying the cost, showing that dimension alone does not describe the utility loss. The construction assumes that the uncertainty directions and labels are supplied to the learner; its exact attainer therefore does not estimate them.

These results have a narrower scope than general fairness-constrained classification. Earlier work supplies barriers for arbitrary marginals (Lechner et al., 2021), exact-fair regression phenomena (Fukuchi & Sakuma, 2022), and finite-sample minimax analyses under nonparametric assumptions (Hou & Zhang, 2024; Zeng et al., 2024). The distinction developed in this paper is the combination of an explicit finite-experiment kernel criterion, a central utility quantile under uniform demographic parity, and an exact multigroup cost under prescribed affine geometry. The supporting oracle comparison is a direct constrained-estimation calculation, consistent with the general reductions perspective. Four synthetic studies quantify the behavior of these particular constructions and the sensitivity of the geometry assumption.

2 Related Work

2.1 Population optimization and statistical fairness

Population fairness optimization and finite-data fairness guarantees answer different questions. Population post-processing, including equality-of-opportunity formulations, provides an important starting point for designing constrained predictors (Hardt et al., 2016). Work on the cost of fairness studies the interaction between loss and fairness constraints (Menon & Williamson, 2018); optimized score transformations provide another algorithmic perspective (Wei et al., 2021). Our population oracle is a benchmark supplied with the true distribution. The learned rule is supplied only with the finite sample and the specified model, so an optimal population feasible point need not be uniformly attainable.

Woodworth et al. (2017) study learning non-discriminatory predictors under equalized-odds constraints. Their finite-sample discussion and Lemmas 4–5 expose the role of approximate constraints in statistical

learning. The qualitative exact-versus-approximate obstacle therefore precedes the present analysis. Our constraint is demographic parity in an explicit finite family, and our central quantity is a high-probability improvement over constant prediction. The common-kernel statement additionally quantifies over arbitrary Borel output laws in finite experiments.

The reductions analysis of Agarwal et al. (2018), particularly Theorems 2–3 and Assumption 1, combines empirical optimization with population constraint and objective control through a complexity-dependent sampling term. Its randomized hypothesis mixtures address a substantially broader learning problem. Proposition 5 specializes the familiar concentration-and-tightening mechanism to four prediction coordinates, with an explicit radius and a simultaneous event. Related algorithmic approaches include fairness-constrained empirical risk minimization (Donini et al., 2018), constrained model training (Cotter et al., 2019), and fairness constraints in classification (Zafar et al., 2019).

FaiREE develops finite-sample post-processing with distribution-free fairness guarantees (Li et al., 2023). Its demographic-parity development in Appendix A.8.1, including Proposition A.11 and Theorems A.12–A.13, works with positive tolerance and threshold-based constructions, together with sample-size and utility conditions. Our shrinkage rule is explicit for a known binary label map and is defined for every nonnegative tolerance. The converse bounds a central utility quantile over all uniformly fair learners in that finite model. The additional converse structure comes with a much more restrictive statistical experiment.

2.2 Impossibility, robustness, and minimax criteria

Lechner et al. (2021) show, in their Claim 1 and Corollary 1, how arbitrary marginal distributions obstruct demographic parity for fixed nonconstant deterministic predictors. We instead prescribe an affine or polynomial population family and allow data-dependent randomized learners. Restricting the uncertainty family can retain nonconstant universally fair rules. The multigroup kernel formula describes exactly which directions survive that restriction, while the probability characterization accounts for the learner’s permissible failure mass.

The robustness analysis of Agarwal et al. (2025) distinguishes the discontinuity of deterministic fair optima in a four-cell construction from a Lipschitz property of randomized optimal values. That comparison concerns population optimization as the distribution changes. Our finite-data output law must meet fairness at every unknown member of a family; continuity of a population optimal value alone does not provide that uniform statistical feasibility. Randomized prediction can improve the population problem while a common-kernel restriction still limits uniformly exactly fair learning.

Fukuchi & Sakuma (2022) analyze demographic-parity-constrained regression in a Gaussian linear model. Their Theorem 1 connects exact-fair learning difficulties with estimation and describes rates under bounded model parameters and a Wasserstein fairness criterion. Thus exact-fair statistical barriers and sampling-scale relaxations occur beyond finite classification. Theorem 1 assumes a finite observation alphabet and polynomial constraints on prediction coordinates, so its proof does not subsume that continuous experiment. Conversely, the central classification quantile used by our envelope differs from a regression error criterion.

Recent minimax analyses further sharpen the distinction between fairness validity and utility. Hou & Zhang (2024) study finite-sample, high-probability fairness and excess risk under nonparametric structure, including strong-density, smoothness, margin, and detection conditions; their Theorems 1 and 3 and Appendix P address quantitative behavior and information restrictions. Zeng et al. (2024) use fairness-aware excess risk in nonparametric demographic-parity classification, with assumptions and rate statements developed around Definitions 4.1 and 4.6, Theorem 4.7, Corollary 6.5, and Theorem 6.7. Those analyses involve unknown regression functions and richer distribution classes. Our known-label experiment instead isolates the price of estimating group frequencies, and its objective is a central high-probability utility threshold. A matching order in this criterion should not be read as a general minimax rate for constrained classification.

Certification also has its own information requirements. Theorems 2.1–2.2 of Gitiaux & Rangwala (2021) concern certification for a fixed representation through information-theoretic conditions and lower and upper statistical bounds. Our learner chooses the predictor, and usefulness of that chosen predictor is the event whose probability is bounded. Related questions arise when sensitive attributes are unavailable (Kallus et al.,

2019), when training data are corrupted (Konstantinov & Lampert, 2022), and when protected attributes are noisy (Celis et al., 2021). These settings motivate robustness questions, but the present experiment observes group membership and uses an uncorrupted sample.

2.3 Multigroup structure and the role of randomization

Zhao & Gordon (2022) derive population accuracy–fairness tradeoffs for multiple groups, including total-variation and barycenter formulations in Theorem 8, Corollary 9, and Theorem 12. In our central multigroup distribution, all groups have the same feature and label laws, so a population-only label-distribution obstruction vanishes: the known label rule is perfectly accurate and fair. The positive cost in Theorem 2 instead comes from requiring one learning procedure to satisfy exact fairness uniformly over independently varying nearby group distributions. The common-kernel approximation cost therefore measures a statistical uniformity restriction rather than a disparity already present at the center.

Agarwal & Deshpande (2022) analyze the power of randomized population classification and representation; Theorem 1 and Proposition 3.6 describe optimal randomized rules through a common acceptance-rate construction. In our model, prediction probabilities already represent this type of randomization. A second randomization chooses between complete output rules and uses the allowed learner failure probability. The exact factor $1 - \delta$ in the multigroup expected-risk result arises from this second layer. It is not an assertion that a population optimizer must deliberately violate fairness with probability δ .

The broader literature also addresses multiclass demographic parity and its optimal classifiers (Denis et al., 2024), Wasserstein-barycenter methods for non-decomposable performance measures (Gaucher et al., 2023), decoupled classification (Dwork et al., 2018), fair regression through reductions (Agarwal et al., 2019), and fairness over rich subgroup collections (Kearns et al., 2018). Our groups and finite prediction coordinates are fixed in advance. Uncertainty-aware comparison of fairness and performance metrics is studied by Barrainkua et al. (2024). The omitted-direction experiment instead concerns misspecification of a supplied geometric model.

3 Problem Setup

3.1 Learners and population guarantees

Let $\Theta \subset \mathbb{R}^k$ be nonempty, open, and connected, with $k \geq 1$. A finite training experiment takes values in a finite set $\mathcal{S}$, with probabilities $p_\theta(s)$ continuous in θ for every s . A nonempty compact set $Q \subset \mathbb{R}^d$ contains the admissible prediction vectors. A learner consists of one Borel probability measure κ_s on Q for each possible observation s . These measures may depend on the known model and tuning parameters, but not on the unknown θ . Its output law is

$$\mu_\theta(D) = \sum_{s \in \mathcal{S}} p_\theta(s) \kappa_s(D), \quad D \subset Q \text{ Borel}.$$

This definition includes randomized algorithms with continuous output distributions. The probability notation $\Pr_\theta$ includes the training sample and learner randomness. A coordinate of q is itself an acceptance probability, so population risk and disparity integrate the subsequent prediction randomization.

Let $M(\theta)$ be a matrix with finitely many rows and d columns, whose entries are real polynomials in θ . Exact fairness means $M(\theta)q = 0$. Write $\Delta_\theta(q)$ for a nonnegative disparity with this zero set; in the binary model below it is an absolute difference. A learner is uniformly (ϵ, δ) -fair when

$$\Pr_\theta\{\Delta_\theta(q) \leq \epsilon\} \geq 1 - \delta \quad \text{for every } \theta \in \Theta.$$

The statement permits a different exceptional training event at each population. It does not require a single event simultaneously measurable under every population law. Nor is it a bound on disparity after averaging the signed residual over output rules.

3.2 The binary frequency experiment

For the quantitative results, take $\Theta = (-1/2, 1/2)^2$, write $\theta = (u, v)$, and observe $n \geq 1$ independent samples from each group, with all $2n$ observations mutually independent. Conditional on group $A = 0$, X is Bernoulli with mean $1/2 + u$; conditional on $A = 1$, its mean is $1/2 + v$. The known label map is $Y = X$. Test groups have probability $1/2$ each. Only u, v are unknown.

A prediction vector $q = (q_{00}, q_{01}, q_{10}, q_{11}) \in [0, 1]^4$ specifies $\Pr(\hat{Y} = 1 \mid A = a, X = x) = q_{ax}$. Define

$$s_0 = q_{01} - q_{00}, \quad s_1 = q_{11} - q_{10}, \quad b = \frac{q_{00} + q_{01} - q_{10} - q_{11}}{2}.$$

The signed acceptance difference, its absolute value, and the misclassification risk are

$$F_\theta(q) = b + us_0 - vs_1, \quad \Delta_\theta(q) = |F_\theta(q)|, \quad (1)$$

$$R_\theta(q) = \frac{1}{2} \left[\left(\frac{1}{2} - u \right) q_{00} + \left(\frac{1}{2} + u \right) (1 - q_{01}) + \left(\frac{1}{2} - v \right) q_{10} + \left(\frac{1}{2} + v \right) (1 - q_{11}) \right]. \quad (2)$$

At the center, $R_0(q) = 1/2 - (s_0 + s_1)/4$. Every four-coordinate constant vector has central risk $1/2$. The exact-fair population oracle is

$$R_\theta^* = \min\{R_\theta(q) : q \in [0, 1]^4, F_\theta(q) = 0\}.$$

The feasible set is nonempty and compact, so the minimum exists. The oracle knows θ ; the learner does not. At the center the vector $(0, 1, 0, 1)$ is feasible and has zero risk.

Write $\hat{u}, \hat{v}$ for the two empirical frequencies minus $1/2$. Counts are sufficient for all constructions, but the learner class includes arbitrary functions of the ordered finite sample. For $a \in (0, 1)$, define $k_a = \sqrt{\log(4/a)/2}$ and $r_a = k_a/\sqrt{n}$. These radii refer to simultaneous estimation of the two frequencies. The multigroup model will specify its own known geometry, label map, group weights, and sample sizes before its theorem.

4 Exact Fairness and the Common Kernel

Exact equality imposes a geometric restriction that persists at every finite sample size. The relevant set is the *common kernel*

$$\mathcal{K} = \{q \in Q : M(\theta)q = 0 \text{ for every } \theta \in \Theta\}.$$

The intersection is taken within the admissible prediction set; Q need not itself be a vector space. A rule in $\mathcal{K}$ is fair regardless of the unknown parameter. The issue is whether a learner can instead select rules that are fair only at the population that generated its sample, while retaining a uniform exact-fairness guarantee.

Theorem 1 (Common-kernel characterization). *Consider the finite experiment, compact prediction set, fixed Borel learner kernel, and polynomial constraint matrix specified in Section 3. For every $\delta \in [0, 1)$, a fixed learner is uniformly $(0, \delta)$ -fair if and only if*

$$\Pr_\theta(q \in \mathcal{K}) \geq 1 - \delta \quad \text{for every } \theta \in \Theta.$$

For Lebesgue-almost every $\theta \in \Theta$, its probability of an exactly fair output outside $\mathcal{K}$ is zero; this assertion holds for any fixed learner, without assuming fairness.

In the binary frequency experiment, $\mathcal{K} = \{(t, t, t, t) : 0 \leq t \leq 1\}$. Consequently every uniformly $(0, \delta)$ -fair learner satisfies $\Pr_0(R_0(q) = 1/2) \geq 1 - \delta$, while $R_0^ = 0$.*

A complete proof, including the polynomial auxiliary lemma and the randomization argument, appears in Appendix A.

Why an output-specific equality is insufficient. Fix $q \notin \mathcal{K}$. At least one row of $M(\theta)q$ is a nonzero polynomial in θ , so its zero set has Lebesgue measure zero. This observation alone does not prove the theorem: the learner may have an uncountable output support, and an uncountable union of null sets need

not be null. The argument instead fixes each observation s , integrates against its entire output measure κ_s , and applies Fubini to the nonnegative Borel indicator. On each bounded parameter box portion B ,

$$\int_B \int_Q \mathbf{1}\{q \notin \mathcal{K}, M(\theta)q = 0\} \kappa_s(dq) d\theta = 0. \quad (3)$$

Finite observation support and a countable exhaustion of Θ then give one null parameter set outside which every observation-conditioned integral vanishes. Weighting by $p_\theta(s)$ proves the almost-everywhere assertion.

The second step is statistical rather than algebraic. The probability of membership in the fixed common kernel is

$$f(\theta) = \mu_\theta(\mathcal{K}) = \sum_{s \in \mathcal{S}} p_\theta(s) \kappa_s(\mathcal{K}). \quad (4)$$

This is continuous because the sum is finite and the learner measures do not depend on θ . Uniform fairness and (3) imply $f \geq 1 - \delta$ almost everywhere. If the inequality failed at any point, continuity would make it fail on a nonempty open ball, contradicting the almost-everywhere bound. Finally, membership in $\mathcal{K}$ implies fairness at every population, which proves the converse implication.

The binary consequence and its scope. The polynomial $b + us_0 - vs_1$ vanishes on the whole open square exactly when its three coefficients vanish. Thus $s_0 = s_1 = b = 0$, forcing all four prediction probabilities to be equal. At the center such rules lose all feature-dependent improvement even though the label rule is fair. The phenomenon holds for every finite n , including $n = 1$, and for arbitrary learner randomization.

The restriction comes from varying u and v independently. If the population family were restricted to the line $u = v$, the perfect label rule would lie in the common kernel. Similarly, the multigroup family below retains nonconstant rules whenever its uncertainty span leaves suitable directions available. This family dependence distinguishes the characterization from impossibility statements over arbitrary marginals (Lechner et al., 2021). Theorem 1 permits zero observation probabilities and does not divide by a likelihood. Its continuity argument concerns the mass of a fixed set, not continuity of the population-specific feasible set or of an optimal rule.

5 Multigroup Kernels and Label Cost

The constant-only kernel in the binary example is an extreme consequence of its uncertainty directions. A richer feature alphabet allows nonconstant prediction rules to be orthogonal to every uncertain direction. We now compute both the surviving dimension and the central label cost, separating the size of the kernel from its alignment with the labels.

5.1 Known affine uncertainty

Let there be $g \geq 2$ observed groups and $m \geq 2$ feature values. The learner knows a strictly positive probability vector $p \in \mathbb{R}^m$, a full-column-rank matrix $B \in \mathbb{R}^{m \times r}$ with $1 \leq r \leq m - 1$ and $B^\top \mathbf{1} = 0$, and a number $\rho > 0$. For group a ,

$$p_a = p + B\theta_a, \quad \theta_a \in (-\rho, \rho)^r.$$

Every coordinate of every such vector is assumed strictly positive. The parameters $\theta_1, \dots, \theta_g$ vary independently over the product of these open cubes. The sum constraint on B ensures that each p_a sums to one.

The common binary label map $y \in \{0, 1\}^m$ is known, with $Y = y_X$. Known test weights $w_a > 0$ sum to one. Group a supplies a known fixed positive integer n_a of independent observations; observations are mutually independent across groups. Only the parameter vectors are unknown. The learner is a fixed Borel kernel into $[0, 1]^{g^m}$, with group blocks q_a . Demographic parity requires $p_a^\top q_a = p_1^\top q_1$ for every a . Population risk is $\sum_a w_a \sum_j p_{a,j} |q_{a,j} - y_j|$.

Theorem 2 (Multigroup kernel and exact central cost). *Under the preceding known affine model, define*

$$K_B = \{q \in [0, 1]^{gm} : B^\top q_a = 0 \ \forall a, \quad p^\top q_a = p^\top q_1 \ \forall a\},$$

$$L(p, B, y) = \min_{\substack{z \in [0, 1]^m \\ B^\top z = 0}} \sum_{j=1}^m p_j |z_j - y_j|.$$

The common kernel is K_B , with affine dimension $g(m - r - 1) + 1$. For $\delta \in [0, 1)$, a learner is uniformly exactly fair with failure probability at most δ if and only if $\Pr_\theta(q \in K_B) \geq 1 - \delta$ for every θ . Every such learner satisfies

$$\Pr_0\{R_0(q) \geq L(p, B, y)\} \geq 1 - \delta, \quad \inf_{\text{uniformly } (0, \delta)\text{-fair learners}} \mathbb{E}_0 R_0(q) = (1 - \delta)L(p, B, y),$$

and the infimum is attained.

For $m = 4$, $p = \mathbf{1}/4$, $y = (0, 1, 0, 1)$, $B_\lambda = (1, -1, \lambda, -\lambda)^\top$, $\lambda \in [0, 1]$, and $\rho = 1/4$, the kernel dimension is $2g + 1$ and $L(p, B_\lambda, y) = (1 + \lambda)/4$. The central exact-fair population oracle has risk zero.

Appendix E proves all assertions, verifies the finite-experiment reduction, and constructs the attainer.

5.2 Coefficient constraints and label alignment

For $a \geq 2$, the acceptance difference is

$$F_a(\theta, q) = p^\top (q_a - q_1) + \theta_a^\top B^\top q_a - \theta_1^\top B^\top q_1. \quad (5)$$

Because the group parameters range independently over an open product, all coefficients must vanish for universal equality. This yields the stated kernel. With $V = \ker B^\top$, full column rank gives $\dim V = m - r$. The $g - 1$ mean-equality conditions have independent restrictions on V^g , since $\mathbf{1} \in V$ and $p^\top \mathbf{1} = 1$. Rank-nullity therefore gives $g(m - r) - (g - 1)$. The point with every coordinate $1/2$ is interior to the prediction cube, so intersection with the cube preserves this affine dimension.

At the center every group has law p . Each group block of a kernel rule incurs label cost at least L , and the positive weights sum to one. Copying a minimizer z^* into all groups attains L and satisfies the mean-equality constraints. The statistical kernel characterization then gives

$$\mathbb{E}_0 R_0(q) \geq L \Pr_0(q \in K_B) \geq (1 - \delta)L. \quad (6)$$

To attain this value, the learner outputs the copied z^* with probability $1 - \delta$ and the copied label vector y with probability δ . The first branch is fair throughout the family, and the second has zero risk at the center. The supplied p, B, y make this a parameter-independent admissible learner; it does not estimate a classifier from the observations. This also explains why the attainable expected cost depends on the allowed failure probability.

The four-cell family isolates an orientation effect. Setting $d_1 = z_2 - z_1$ and $d_2 = z_4 - z_3$, its kernel constraint is $d_1 + \lambda d_2 = 0$, while the central loss is

$$\frac{1}{2} - \frac{d_1 + d_2}{4} = \frac{1}{2} - \frac{1 - \lambda}{4} d_2 \geq \frac{1 + \lambda}{4}.$$

Equality is attained at $z^* = ((1 + \lambda)/2, (1 - \lambda)/2, 0, 1)$. Thus the cost varies from $1/4$ to $1/2$ without any change in rank or kernel dimension. The relevant geometry is alignment with the known labels.

At the center, all group label distributions coincide and the label predictor is population fair. The positive cost is consequently distinct from a population label-distribution mismatch, as studied by Zhao & Gordon (2022). It results from requiring exact fairness over a neighborhood of independently varying group laws. Knowledge of that neighborhood is binding: a rule orthogonal to the columns of B need not be orthogonal to an omitted direction. The sensitivity experiment in Section 8 tests precisely this boundary.

6 Utility at Positive Tolerance

A positive tolerance replaces exact polynomial zero sets by sets of nonzero width. The width must be compared with the resolution of the training experiment. The binary frequency model makes both quantities explicit: useful prediction forces a slope, while nearby Bernoulli product experiments differ in total variation by at most a constant times $\sqrt{n}$ times their parameter distance.

6.1 A local probability bound

Theorem 3 (Useful-output probability). *In the binary frequency experiment, let $n \geq 1$, $\epsilon \geq 0$, and $\delta \in [0, 1]$. Every uniformly (ϵ, δ) -fair learner satisfies, for $\gamma \in (0, 1/2]$, $h \in (0, 1/4]$, and $\epsilon < 2\gamma h$,*

$$\Pr_0\{R_0(q) \leq 1/2 - \gamma\} \leq \min \left\{ 1, \frac{\delta + \sqrt{8n/3}h}{1 - \epsilon/(2\gamma h)} \right\}.$$

For any sequence of such learners with a fixed δ , if $\epsilon_n \sqrt{n} \rightarrow 0$, then for every fixed $\gamma \in (0, 1/2]$,

$$\limsup_{n \rightarrow \infty} \Pr_0\{R_0(q) \leq 1/2 - \gamma\} \leq \delta.$$

Appendix B provides a complete derivation, including the finite-space total-variation inequality and the zero-tolerance sequence case.

To expose the mechanism, let $A_\gamma = \{q : R_0(q) \leq 1/2 - \gamma\}$ and let ν_h be uniform on $[-h, h]^2$. For $q \in A_\gamma$, the sum of the two slopes is at least 4γ , so at least one slope is at least 2γ . Holding the other parameter coordinate fixed, the set where $|b + us_0 - vs_1| \leq \epsilon$ has interval length at most ϵ/γ . Dividing by the side length $2h$ and integrating yields

$$\nu_h\{\theta : |F_\theta(q)| \leq \epsilon\} \leq \frac{\epsilon}{2\gamma h}, \quad q \in A_\gamma. \quad (7)$$

This is the geometric reason that a useful fixed output cannot be fair over most nearby parameters when ϵ is small.

Let P_θ denote the full training law. Independence and the Bernoulli relative-entropy identity give

$$D(P_0 \| P_\theta) = -\frac{n}{2} \{\log(1 - 4u^2) + \log(1 - 4v^2)\} \leq \frac{16}{3}nh^2, \quad \text{TV}(P_0, P_\theta) \leq \sqrt{\frac{8}{3}}\sqrt{n}h. \quad (8)$$

The restriction $h \leq 1/4$ controls the logarithmic denominator. Applying the same fixed learner kernel cannot increase the difference in probabilities of any output event.

The final step fixes the central output law μ_0 before averaging. Uniform fairness bounds the probability of A_γ intersected with the unfair-output set under each μ_θ by δ . Equation (8) transfers that bound to μ_0 . Equation (7), integrated against this fixed measure, lower-bounds the same averaged probability by $(1 - \epsilon/(2\gamma h))\mu_0(A_\gamma)$. Rearranging gives the theorem. Both the strict denominator condition and the local radius matter; optimizing a displayed bound outside their ranges would be invalid.

For $\epsilon_n \sqrt{n} \rightarrow 0$, a radius can be chosen so that both $\sqrt{n}h_n$ and ϵ_n/h_n vanish. The result then allows useful outputs only within asymptotic probability δ . This conclusion is about a fixed nonzero utility gain. It does not identify optimal constants for shrinking gains, and the finite-radius probability bound can be vacuous at positive tolerances.

6.2 The central utility envelope

To quantify shrinking gains, fix $\delta, \eta \in (0, 1)$ with $\delta + \eta < 1$ and define

$$G_n(\epsilon; \delta, \eta) = \sup \left\{ \gamma \in [0, 1/2] : \begin{array}{l} \text{some uniformly } (\epsilon, \delta)\text{-fair learner has} \\ \Pr_0(R_0(q) \leq 1/2 - \gamma) \geq 1 - \eta \end{array} \right\}.$$

The feasible set contains zero because the constant rule is uniformly exactly fair. Fairness confidence $1 - \delta$ and utility confidence $1 - \eta$ serve separate requirements.

Theorem 4 (Finite-sample central utility). *In the binary frequency experiment, for $n \geq 2$, $\epsilon \geq 0$, and the preceding confidence parameters,*

$$\min \left\{ \frac{1}{2}, \frac{\epsilon \sqrt{n}}{4(k_\delta + k_\eta)} \right\} \leq G_n(\epsilon; \delta, \eta) \leq \min \left\{ \frac{1}{2}, \frac{2\sqrt{8/3}(1-\eta)}{(1-\eta-\delta)^2} \epsilon \sqrt{n} \right\}.$$

Thus, at fixed permitted confidence levels, G_n has matching order $\min\{1, \epsilon\sqrt{n}\}$, and $G_n(0; \delta, \eta) = 0$.

The full construction, concentration proof, converse branches, and supremum argument appear in Appendix D.

The attainable side uses a common shrinkage of the known label rule:

$$s = \min \left\{ 1, \frac{\epsilon}{|\hat{u} - \hat{v}| + 2r_\delta} \right\}, \quad q^{\text{sh}} = \left(\frac{1-s}{2}, \frac{1+s}{2}, \frac{1-s}{2}, \frac{1+s}{2} \right). \quad (9)$$

Its risk is $(1-s)/2$ at every population and its disparity is $s|u-v|$. With probability at least $1 - \delta$, both frequencies are within r_δ of their means, so the denominator controls $|u-v|$ and implies fairness. At the center, a separate frequency event of probability at least $1 - \eta$ gives $|\hat{u} - \hat{v}| \leq 2r_\eta$. Substitution yields the lower bound. The denominator is positive even when the empirical frequencies agree, and $\epsilon = 0$ produces the constant rule.

For the converse, set $c = \sqrt{8/3}$, $a = 1 - \eta - \delta > 0$, and $h = a/(2c\sqrt{n})$. The condition $n \geq 2$ makes this an admissible radius. If an attainable $\gamma > 0$ satisfies $\epsilon < 2\gamma h$, Theorem 3 implies

$$(1-\eta) \left(1 - \frac{\epsilon}{2\gamma h} \right) \leq \delta + \frac{a}{2} \implies \gamma \leq \frac{(1-\eta)\epsilon}{ah} = \frac{2c(1-\eta)}{a^2} \epsilon \sqrt{n}. \quad (10)$$

In the complementary branch, $\gamma \leq \epsilon/(2h)$ gives an at least as strong bound. This bounds each attainable gain without assuming an optimizing learner exists; taking the supremum and using the risk cap $1/2$ completes the result.

The theorem separates three regimes. Below the sampling scale, the central gain vanishes proportionally to $\epsilon\sqrt{n}$. At a tolerance of order $n^{-1/2}$, a constant-order central gain is attainable. For sufficiently large scaled tolerance, the lower bound reaches the maximal gain $1/2$. This last statement is still a central high-probability guarantee, accompanied by uniform fairness, rather than a global excess-risk theorem.

The constants deteriorate as $\delta + \eta$ approaches one and are not optimized. The two confidence events in the construction need not have a common probability $1 - \min\{\delta, \eta\}$; no such intersection claim is used. The model's known labels also remove regression-function estimation. These facts distinguish the result from broader nonparametric risk rates (Hou & Zhang, 2024; Zeng et al., 2024). Its role is to make the exact-to-approximate transition explicit for one fully specified statistical experiment.

7 Comparison with the Fair Oracle

The central envelope measures improvement at one law. An oracle inequality asks a different question: can a learned rule compete with the best exactly fair predictor at the actual population? In the binary experiment, a standard empirical constraint-tightening argument gives a simultaneous fairness and one-sided risk guarantee once the tolerance accommodates estimation error. This comparison also provides a useful baseline for the shrinkage construction.

Fix $\delta \in (0, 1)$, set $r = r_\delta$, and define $\hat{F} = F_{\hat{u}, \hat{v}}$ and $\hat{R} = R_{\hat{u}, \hat{v}}$ by substitution into the formulas in Section 3. For $\epsilon \geq 4r$, choose a minimizer

$$\hat{q} \in \arg \min_{q \in [0, 1]^4} \{ \hat{R}(q) : |\hat{F}(q)| \leq \epsilon - 2r \},$$

using a fixed deterministic tie rule. For $\epsilon < 4r$, define the procedure to return the four-coordinate constant $1/2$. The feasible set in the displayed program contains constants and is compact. A fixed lexicographic minimizer exists, and finite observation support makes the resulting rule measurable.

Proposition 5 (Comparison with the exact-fair oracle). *In the binary frequency experiment, for $n \geq 1$, $\delta \in (0, 1)$, and $\epsilon \geq 4r_\delta$, the tightened empirical program satisfies, for every θ ,*

$$\Pr_{\theta}\{\Delta_{\theta}(\hat{q}) \leq \epsilon, R_{\theta}(\hat{q}) \leq R_{\theta}^* + 2r_{\delta}\} \geq 1 - \delta.$$

At fixed δ , if $\epsilon_n \sqrt{n} \rightarrow \infty$, the inequality applies eventually and its additive risk term tends to zero.

Appendix C gives the concentration derivation, existence and tie-selection argument, and exact event accounting. The fallback is exactly fair, but the displayed oracle comparison is asserted only in the stated $\epsilon \geq 4r_\delta$ range.

The mechanism uses one event $E = \{|\hat{u} - u| \leq r, |\hat{v} - v| \leq r\}$, with probability at least $1 - \delta$. Bounded prediction coordinates give the uniform perturbation bounds

$$\sup_q |\hat{F}(q) - F_{\theta}(q)| \leq 2r, \quad \sup_q |\hat{R}(q) - R_{\theta}(q)| \leq r \quad \text{on } E. \quad (11)$$

For an exact-fair population minimizer q_{θ}^* , this event gives $|\hat{F}(q_{\theta}^*)| \leq 2r \leq \epsilon - 2r$. The oracle is therefore feasible for the empirical program. Empirical optimality then yields

$$R_{\theta}(\hat{q}) \leq \hat{R}(\hat{q}) + r \leq \hat{R}(q_{\theta}^*) + r \leq R_{\theta}^* + 2r. \quad (12)$$

The same event and empirical feasibility give $|F_{\theta}(\hat{q})| \leq |\hat{F}(\hat{q})| + 2r \leq \epsilon$.

The concentration, tightening, and comparison chain are standard components of constrained learning, consistent with the general framework of Agarwal et al. (2018). Their explicit specialization clarifies the benchmark used in the experiments. Since the learned rule may use positive disparity while the comparator must be exactly fair, $R_{\theta}(\hat{q}) - R_{\theta}^*$ can be negative. The proposition bounds this signed excess from above. Neither absolute convergence to the oracle risk nor a bound on every realized output follows from the statement.

8 Experiments

The experiments test the quantitative behavior of the specified finite models. They use exact population risk and disparity, so evaluation has no test-set sampling error. Training samples and learner randomization are repeated independently within each cell. The studies address four questions: whether central utility follows the scaled tolerance, how the tightened program behaves relative to its oracle bound, whether fixed kernel rank determines label cost, and how an omitted uncertainty direction affects equality. Complete grids, seed schedules, numerical tolerances, and uncertainty calculations appear in Appendix F.

8.1 Central utility across tolerance and sample size

For the shrinkage rule in (9), we set $\delta = \eta = 0.1$, vary n over 32, 128, 512, 2048, 8192, and vary $t = \epsilon \sqrt{n}$ over 0, 0.03, 0.1, 0.3, 1, 3, 10. Each cell uses 500 independent training samples. We evaluate the center and three off-center population pairs, including unequal group frequencies and parameters near the boundary. The central statistic is the empirical lower tenth percentile of the gain $1/2 - R_0(q)$. It describes this learner rather than directly estimating the supremum G_n .

Figure 1 compares the empirical quantile with the attainable and converse envelope bounds. At $t = 1$, the observed central lower-quantile gain is approximately 0.1294 for both $n = 128$ and $n = 8192$; the analytic attainable lower bound is approximately 0.0920. At $t = 0.1$, the corresponding gains are approximately 0.01294 and 0.01290. These values illustrate the dependence on scaled tolerance, with substantial room between the conservative theoretical constants. The gain is zero at zero tolerance and reaches $1/2$ on the tested $t = 10$ cells.

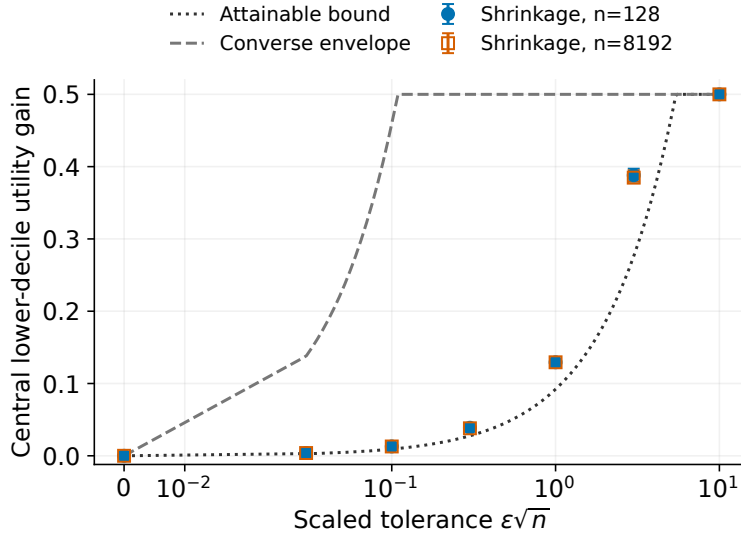


Figure 1: Central utility of the shrinkage learner versus scaled tolerance, with $\delta = \eta = 0.1$. Points show the empirical lower 10th percentile over 500 independent training trials per point; bars are pointwise 95% percentile-bootstrap intervals from 400 resamples, sometimes smaller than the markers. Empirical points are not interpolated. The dotted attainable bound and dashed converse envelope bound optimal central quantile utility, using the local probability inequality for the converse. Near-coincident points at $n = 128$ and 8192 illustrate the predicted scale; separation from the analytic bounds shows conservative constants.

All 70,000 shrinkage outputs across the 140 parameter–sample-size–tolerance cells met the population fairness tolerance. Each 500-trial all-success cell has a pointwise 95% Wilson lower confidence limit of approximately 0.9924, above the required 0.9. This is empirical calibration over the tested grid, not verification at every parameter. We also evaluated the finite-radius bound in Theorem 3 at gain threshold 0.1. It is capped at one throughout the tested positive-tolerance grid and is consequently uninformative there as a numerical upper bound. The scale characterization should therefore be distinguished from sharp probability constants.

8.2 Signed excess risk and the joint oracle event

The second study uses $n = 128, 512, 2048, 8192$, the same four population pairs, 200 trials per cell, and $\epsilon = 4r_{0.1}$. We compare the tightened empirical program, empirical equality without tightening, constant prediction, and the known perfect label rule. A population linear program supplies the exact-fair oracle benchmark.

Figure 2 reports mean signed excess risk at $(u, v) = (0.15, -0.15)$. For the tightened program it changes from approximately -0.1168 at $n = 128$ to -0.01551 at $n = 8192$. The negative values are consistent with the comparison: the learned rule may exploit a positive tolerance, while the oracle is constrained to exact equality. The plotted $2r_\delta$ curve is the theorem’s per-output high-probability bound; it is not a bound specifically on the mean statistic.

The direct check of Proposition 5 is the joint event of population fairness and signed excess at most $2r_\delta$. All 3,200 tightened-program trials met that event. Each 200-trial cell has a pointwise Wilson lower limit of approximately 0.9812. At the displayed unequal-frequency population, the perfect predictor has signed excess -0.15 , but fails the fairness tolerance from $n = 512$ onward. Low risk alone therefore does not establish a valid comparison. The empirical-equality baseline has signed excess near zero on average, while its population constraint remains a separate quantity.

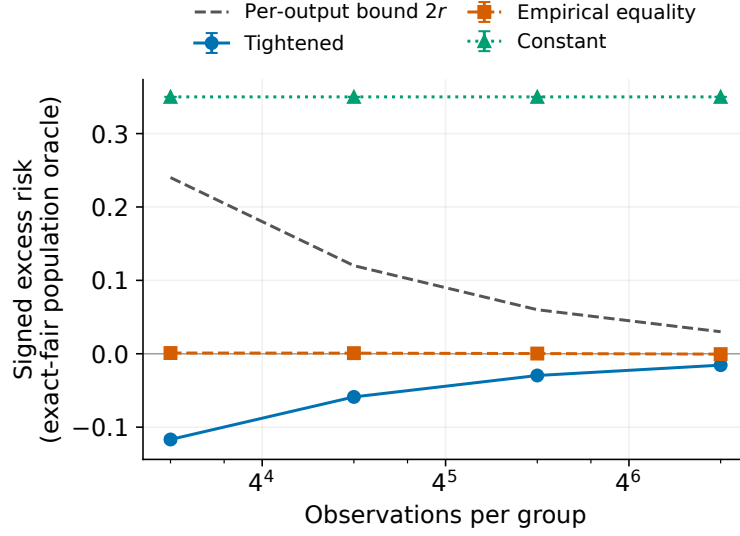


Figure 2: Signed excess risk over the exact-fair population oracle at $(u, v) = (0.15, -0.15)$ and tolerance $\epsilon = 4\sqrt{\log(40)/(2n)}$. Points average 200 independent training trials; bars are pointwise normal 95% mean intervals, some smaller than their markers. The tightened LP can have negative excess because its positive tolerance permits comparison with a stricter exact-fair oracle. The dashed curve is the high-probability per-output upper bound $2r$ for the tightened LP, not a bound on the plotted mean or on absolute error. Empirical equality and a constant rule provide distinct baselines.

8.3 Label cost at fixed kernel dimension

The third study uses three groups, four cells, equal test weights, 128 observations per group, and the B_λ family of Theorem 2. We vary λ over 0, 0.25, 0.5, 0.75, 1, test the center and $\theta = (-0.1, 0, 0.1)$, and use 200 independent trials per cell. Comparators are the copied kernel minimizer, its $\delta = 0.1$ mixture with the perfect rule, a three-group empirical-equality linear program, and the constant rule.

The deterministic kernel risks at the center are 0.25, 0.3125, 0.375, 0.4375, 0.5, while the affine kernel dimension remains seven. The mixture’s expected risks are 0.225, 0.28125, 0.3375, 0.39375, 0.45. The observed means, approximately 0.21875, 0.28438, 0.33938, 0.38719, 0.465, have pointwise mean intervals containing these expectations. The empirical-equality program has much lower central risk, approximately 0.022–0.024, but population equality occurs in at most one of the 200 trials in each central cell. Its low risk is therefore not evidence against the uniformly fair cost.

The kernel and constant controls retain population equality. For the mixture at the off-center parameter, the observed exact-fair proportions range from 0.865 to 0.935, with all pointwise Wilson intervals containing 0.9. At the center, both mixture branches are fair, so its fairness probability is one rather than 0.9. The formula $1 - \delta$ specifies a uniform lower bound, not equality of the fairness probability at every parameter.

8.4 Sensitivity to an omitted direction

The final study perturbs the $\lambda = 0.5$ model by $p_a = p + B\theta_a + \tau z_a C$, with $C = (1, 0, 0, -1)^\top$, $z = (-1, 0, 1)$, and $\theta = (-0.1, 0, 0.1)$. We use $\tau = 0, 0.005, 0.01, 0.02, 0.04$, $n = 128, 512$, and 200 training draws per cell. The nominal kernel rule is $(0.75, 0.25, 0, 1)$, copied across groups. The controls use the expanded known span $[B, C]$, empirical equality, or constant prediction.

The nominal rule’s maximum pairwise population disparity equals $\tau/2$, giving 0.0025 to 0.02 at the positive amplitudes for both sample sizes. The expanded-kernel rule $(1, 0.5, 0, 1)$ retains equality and has risk 0.375, the same as the nominal rule in this symmetric example. Thus this particular correction costs no additional

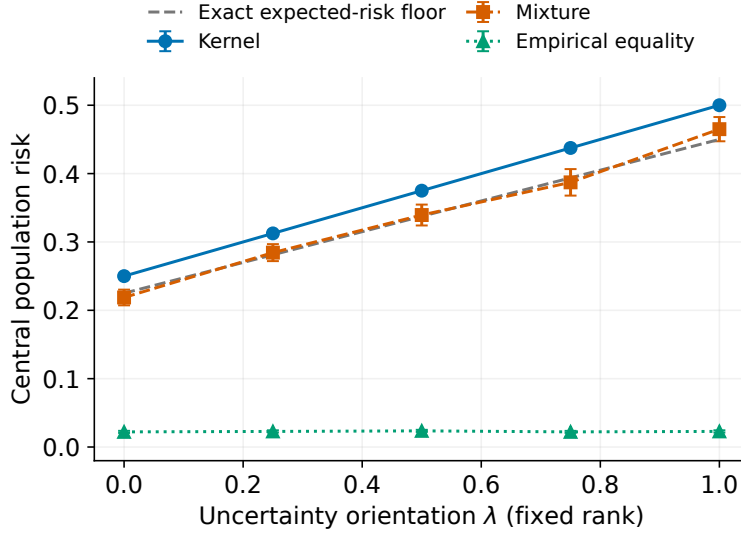


Figure 3: Central risk versus uncertainty orientation for three groups, four categories and 128 observations per group, with fixed kernel dimension seven. Points average 200 independent trials; bars are pointwise normal 95% mean intervals. The kernel rule has risk $(1 + \lambda)/4$, and its 0.1-mixture with perfect prediction attains expected risk $0.9(1 + \lambda)/4$. The empirical-equality LP has lower risk but is not comparably population-fair: center exact-fairness rates were 0–0.005 using numerical tolerance 10^{-8} . The dashed floor applies to uniformly $(0, 0.1)$ -fair learners.

central risk. These two rules are deterministic; repeating their rows does not create statistical evidence. The empirical-equality comparator is stochastic through its training data and fails population equality in all 200 trials per cell. Positive τ lies outside the nominal uncertainty model, so the observed disparity identifies sensitivity of the supplied-geometry assumption rather than a counterexample to the theorem.

9 Limitations

The finite observation alphabet is essential to the stated common-kernel theorem. Its proof forms a finite sum of continuous likelihoods and fixed output probabilities. Extending the result to continuous observations would require additional domination, measurability, and continuity conditions; the finite-experiment argument alone does not supply them. Polynomial constraint dependence is likewise a stated structural assumption, rather than a claim about arbitrary fairness functionals.

The quantitative utility envelope concerns two observed groups, binary features, known deterministic labels, equal test weights, and independent samples. Its fairness condition is uniform over a specified open square, while usefulness is measured only at the center. The matching order is in a lower-quantile improvement over constant prediction, with fixed confidence levels. The constants are conservative and the local probability bound is numerically vacuous on much of the tested grid. Unknown regression functions, label noise, group-blind decisions, and growing feature classes introduce learning problems beyond this experiment.

In the multigroup result, the learner knows the reference law, uncertainty matrix, label map, and test weights. The exact expected-risk attainer can therefore ignore the data. Its value characterizes the price of uniform exact fairness in a prescribed family; it is not an algorithm for discovering a fair representation or an unknown uncertainty span. Independent variation of the group parameters is also consequential. Coupled parameter restrictions can enlarge the common kernel and alter the cost.

The synthetic studies use exact population metrics and a small number of controlled families. Their uncertainty intervals are pointwise Monte Carlo summaries; they do not prove the uniform guarantees or establish

simultaneous coverage over the full grid. The misspecification study tests one omitted direction and assumes that the expanded control is told that direction. Real-data effectiveness, estimation of geometry, and robustness to unmodeled population changes remain separate empirical and mathematical questions.

10 Conclusion

Uniform population fairness creates a distinction between a fair predictor at one distribution and an output rule that finite data can select with a valid guarantee throughout a family. In finite polynomial experiments, the common kernel gives an exact characterization of the latter requirement. For the binary frequency model, positive tolerance restores central utility on the scale $\epsilon\sqrt{n}$, with an explicit shrinkage construction and a local-experiment converse. The multigroup affine model shows that exact fairness can preserve nonconstant rules and that their cost depends on label alignment, even at fixed kernel dimension.

The statistical requirement and the information supplied to the learner jointly determine these conclusions. The central multigroup oracle has zero risk, yet uniform exact fairness imposes a positive cost; supplying a larger uncertainty span can change which rules are feasible. A natural next question is to characterize the best central utility when the uncertainty span and the label probabilities must both be estimated, while retaining a finite-sample population fairness guarantee under controlled geometry error.

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Appendix

A Proof of the Kernel Characterization

Proof. We prove Theorem 1.

Step 1: Polynomial auxiliary lemma. The polynomial zero-set fact is a special case of the analytic result discussed by Mityagin (2015); the required integration theorem is stated in Hunter (2011, Theorem 5.18). We give the polynomial argument explicitly. A nonzero real polynomial in k variables has a Lebesgue-null zero set. For $k = 1$ it has at most its degree many roots, by repeated polynomial division. For $k > 1$, write $p(x, t) = \sum_{j=0}^m a_j(x)t^j$ with at least one nonzero coefficient polynomial a_j in $k - 1$ variables. The set on which all a_j vanish is contained in the zero set of that nonzero coefficient, hence is null by induction. For every other x the section in t has finitely many zeros. On any bounded box the zero-set indicator is Borel, bounded and integrable. Fubini gives zero integral: the exceptional x set has zero measure and every other section has zero one-dimensional measure. Countably exhausting Euclidean space by bounded boxes proves the lemma.

Step 2: Measurability and quantifiers. For fixed θ , $\{q : M(\theta)q = 0\}$ is closed relative to compact Q . Its arbitrary intersection $\mathcal{K}$ is closed, hence Borel. The map $(\theta, q) \mapsto M(\theta)q$ is jointly continuous. Thus $g(\theta, q) = 1_{\{q \notin \mathcal{K}, M(\theta)q=0\}}$ is product Borel. For each fixed $q \notin \mathcal{K}$, some row of $M(\theta)q$ is a polynomial not identically zero, since it is nonzero at at least one parameter by definition of $\mathcal{K}$. Consequently the parameter set where all rows vanish has measure zero by Step 1. For $q \in \mathcal{K}$, g is identically zero. Therefore $\int_B g(\theta, q)d\theta = 0$ for every q and every bounded Borel box portion $B = \Theta \cap [-j, j]^k$.

Step 3: Interchanging learner randomization. Write κ_s for the fixed output probability measure after observation s . For each s , Fubini on $B \times Q$ gives $\int_B \int_Q g(\theta, q)\kappa_s(dq)d\theta = 0$. The inner function is nonnegative, so it vanishes almost everywhere on B . There are finitely many s and countably many B , hence a single null set N outside which every such inner integral vanishes. For $\theta \notin N$, $\Pr_\theta(q \notin \mathcal{K}, M(\theta)q = 0) = \sum_s p_\theta(s) \int g(\theta, q)\kappa_s(dq) = 0$. This is not an uncountable union over outputs; the integration is essential. It establishes the almost-everywhere assertion for every fixed learner, without any fairness assumption.

Step 4: From almost everywhere to every parameter. Set $f(\theta) = \Pr_\theta(q \in \mathcal{K}) = \sum_s p_\theta(s)\kappa_s(\mathcal{K})$. The factors $\kappa_s(\mathcal{K})$ do not depend on θ by the fixed learner-kernel assumption, so the finite sum is continuous. If the learner is uniformly exactly fair, Step 3 implies $f(\theta) \geq 1 - \delta$ off N . If $f(\theta_0) < 1 - \delta$ at any point of the open parameter set, continuity gives an open ball inside Θ where the same strict inequality holds. Every nonempty Euclidean open ball has positive Lebesgue measure, contradicting the almost-everywhere bound. Thus the inequality holds everywhere. Conversely, $q \in \mathcal{K}$ implies exact fairness at each parameter, proving sufficiency. If $\mathcal{K}$ is empty, the necessity says that no learner satisfies a guarantee with $\delta < 1$; no nonemptiness of the kernel was silently assumed.

Step 5: Demographic parity instance. In the two-group binary experiment the finite training likelihoods are products of Bernoulli probabilities and are continuous. The fairness residual is $b + us_0 - vs_1$. For it to vanish on the open parameter square its three coefficients must vanish: $s_0 = s_1 = b = 0$. The first two equalities make each group's two coordinates equal; the last makes their constants equal across groups. Conversely any four-coordinate constant makes the residual zero. At $\theta = 0$ its risk is $1/2$. The predictor $(0, 1, 0, 1)$ has zero risk and zero disparity at this parameter, so the exact fair optimum is zero. Kernel mass at least $1 - \delta$ proves the stated risk conclusion. This argument permits arbitrary continuous algorithmic randomization and independent Bernoulli prediction randomization already represented by q .

Boundary checks. The general argument allows zero sampling probabilities, since Step 3 needs no division by them. It works at $n = 1$ in the instance and for $\delta = 0$. It does not apply a full-dimensional conclusion to the restricted line $u = v$: on that different family the kernel contains the perfect predictor. The guarantee is conditional on the realized predictor; averaging signed disparity over training outcomes would define a different object. $\square$

B Proof of the Local Probability Bound

Proof. We prove Theorem 3.

Step 1: Useful prediction supplies a slope. Let $B_\gamma = \{q : R_0(q) \leq 1/2 - \gamma\}$. The identity $R_0 = 1/2 - (s_0 + s_1)/4$ gives $s_0 + s_1 \geq 4\gamma$ on this event. At least one of s_0, s_1 is at least 2γ , and hence has absolute value at least 2γ . Let ν_h be uniform measure on the parameter square $[-h, h]^2$. For any fixed $q \in B_\gamma$ with $|s_0| \geq 2\gamma$, fix v and solve $|b + us_0 - vs_1| \leq \epsilon$ as an interval in u . Its length before intersection with $[-h, h]$ is $2\epsilon/|s_0|$. After dividing by $2h$, the conditional probability is at most $\epsilon/(h|s_0|) \leq \epsilon/(2\gamma h) =: \alpha$. If instead $|s_1| \geq 2\gamma$, condition on u and integrate v . Fubini therefore gives $\nu_h\{\theta : |F_\theta(q)| \leq \epsilon\} \leq \alpha$ for every $q \in B_\gamma$. At $\epsilon = 0$ the interval has zero length, so the same calculation applies.

Step 2: Finite-space total variation inequality, with derivation. For finite probability vectors P, Q with positive Q , define $D(P\|Q) = \sum_i P_i \log(P_i/Q_i)$ and $\text{TV}(P, Q) = \frac{1}{2} \sum_i |P_i - Q_i|$. For a subset C , convexity of $z \log z$ and Jensen with weights $Q_i/Q(C)$ give $\sum_{i \in C} P_i \log(P_i/Q_i) \geq P(C) \log(P(C)/Q(C))$; the empty or zero-mass cases follow by limits. Choose $C = \{i : P_i \geq Q_i\}$. Writing $a = P(C), c = Q(C)$, we have $a - c = \text{TV}(P, Q)$ and, after applying the same inequality to C^c , $D(P\|Q) \geq d(a|c)$, the binary relative entropy. For $0 < c < 1$, viewed as a function of a , $d(c|c) = 0$, its first derivative at c is zero, and its second derivative is $1/[a(1-a)] \geq 4$. Twice integrating this lower bound, on either side of c , yields $d(a|c) \geq 2(a-c)^2$. Endpoints follow by continuity. Thus $\text{TV}(P, Q) \leq \sqrt{D(P\|Q)/2}$. Also, for any $k_i \in [0, 1]$, $|\sum_i (P_i - Q_i)k_i| \leq \text{TV}(P, Q)$: the positive and negative parts each sum to TV. Taking k_i as the output-kernel probability of an arbitrary output event proves event transfer through randomized learning, without assuming a deterministic learner.

Step 3: Compute the local experiment distance. Let $\mathbb{P}_\theta$ denote the law of the whole stratified training sample. Product independence and expansion of the log likelihood give

$$D(\mathbb{P}_0\|\mathbb{P}_\theta) = n \left[-\frac{1}{2} \log(1 - 4u^2) - \frac{1}{2} \log(1 - 4v^2) \right].$$

Indeed the one-observation term is $\frac{1}{2} \log[(1/2)/(1/2 + u)] + \frac{1}{2} \log[(1/2)/(1/2 - u)]$. For $0 \leq z \leq 1/4$, $-\log(1 - z) = \int_0^z (1 - t)^{-1} dt \leq (4/3)z$. Since $|u|, |v| \leq h \leq 1/4$, each one-group term is at most $(8/3)h^2$. There are two groups with n observations each, giving $D \leq (16/3)nh^2$ and TV at most $\beta = \sqrt{8n/3}h$. The bound may exceed one, but remains a valid upper bound.

Step 4: Fix the output law before averaging. Let μ_0 denote the learner output law under $\mathbb{P}_0$. For each θ define $E_\theta = B_\gamma \cap \{q : |F_\theta(q)| > \epsilon\}$. Uniform fairness gives $\mu_\theta(E_\theta) \leq \delta$. Step 2 applied to this fixed event for each θ gives $\mu_0(E_\theta) \leq \delta + \beta$. Integrating over ν_h and interchanging bounded Borel indicators by Fubini gives

$$\delta + \beta \geq \int \mu_0(E_\theta) d\nu_h(\theta) = \int_{B_\gamma} \nu_h\{|F_\theta(q)| > \epsilon\} d\mu_0(q) \geq (1 - \alpha)\mu_0(B_\gamma).$$

Here μ_0 is independent of the integration parameter. The prescribed strict inequality $\epsilon < 2\gamma h$ makes $1 - \alpha > 0$. Division and the elementary probability bound by one establish the claim.

Step 5: Sequence and zero-tolerance cases. Put $t_n = \epsilon_n \sqrt{n}$. On indices with $t_n > 0$, choose $h_n = \sqrt{t_n}/\sqrt{n}$. If $t_n \rightarrow 0$, eventually $h_n \leq 1/4$ and $\alpha_n = \sqrt{t_n}/(2\gamma) < 1$, while $\beta_n = \sqrt{8/3} \sqrt{t_n} \rightarrow 0$. The displayed bound then tends to δ . On indices where $\epsilon_n = 0$, send h to zero in the finite- n bound to get $\mu_0(B_\gamma) \leq \delta$ directly. Combining the two index sets proves the limsup claim.

Boundary checks. The model at every parameter of the square has probabilities in $[1/4, 3/4]$, so all likelihood divisions are legitimate. At $\gamma = 1/2$ the useful event forces both slopes to be one; the strip bound remains valid. At $n = 1$ all formulas remain finite. At the excluded boundary $\epsilon = 2\gamma h$, the method has no informative denominator and no claim is made. The theorem bounds useful prediction at the central distribution, while its fairness premise is uniform over the full open family; it does not claim a pointwise impossibility for an algorithm given the true parameter. $\square$

C Proof of the Oracle Comparison

Proof. We prove Proposition 5.

Step 1: Derive the needed concentration bound. For $Z \sim \text{Bern}(p)$ set $\psi(\lambda) = \log \mathbb{E} \exp(\lambda(Z - p))$. It satisfies $\psi(0) = \psi'(0) = 0$. Differentiating its finite exponential sum shows that $\psi''(\lambda)$ is the variance of a Bernoulli variable under exponential tilting. If its tilted success probability is t , that variance is $t(1-t) \leq 1/4$. Integration of the second derivative, for positive or negative λ , yields $\psi(\lambda) \leq \lambda^2/8$. Independence gives $\mathbb{E} \exp(\lambda \sum_{i=1}^n (Z_i - p)) \leq \exp(n\lambda^2/8)$. Markov's inequality with $\lambda = 4r > 0$ then yields $\Pr(\bar{Z} - p \geq r) \leq \exp(-2nr^2)$; applying the same argument to the negative sum gives the lower tail. Union bounding the two tails in two groups gives the event

$$\mathcal{E} = \{|\hat{u} - u| \leq r, |\hat{v} - v| \leq r\}, \quad \Pr_{\theta}(\mathcal{E}) \geq 1 - 4e^{-2nr^2} = 1 - \delta.$$

The result also covers deterministic Bernoulli endpoints as an auxiliary calculation, though the stated parameter family is interior.

Step 2: Uniform constraint and objective perturbation. For every $q \in [0, 1]^4$, $|s_a| \leq 1$ and $\hat{F}(q) - F_{\theta}(q) = (\hat{u} - u)s_0 - (\hat{v} - v)s_1$. On $\mathcal{E}$ the absolute difference is at most $2r$ simultaneously for every q . The risk difference is

$$\hat{R}(q) - R_{\theta}(q) = \frac{1}{2}[(\hat{u} - u)(1 - q_{01} - q_{00}) + (\hat{v} - v)(1 - q_{11} - q_{10})].$$

Each parenthesis involving q has absolute value at most one, so the absolute risk difference is at most r , again uniformly. These deterministic inequalities justify using the data-dependent optimizer without a pointwise-to-uniform gap.

Step 3: Existence, selection and comparator feasibility. The cube is compact. Both the population exactly fair feasible set and empirical tightened feasible set are closed; each contains the zero constant predictor because $\epsilon - 2r \geq 0$. A continuous linear risk therefore attains a minimum on each. There are only finitely many training samples at each fixed n . For each sample choose one minimizer, for example by successive minimization of the four coordinates over the compact optimal face. Such a choice is a measurable map on the finite data space; no general measurable-selection theorem is needed. Let q^* attain the population exactly fair optimum. On $\mathcal{E}$, $|\hat{F}(q^*)| \leq 2r \leq \epsilon - 2r$. Thus the exact comparator is feasible for the empirical problem. The second inequality is exactly why the theorem requires $\epsilon \geq 4r$.

Step 4: Simultaneous conclusions. Empirical feasibility and Step 2 give $|F_{\theta}(\hat{q})| \leq |\hat{F}(\hat{q})| + 2r \leq \epsilon$. Empirical optimality and the two uniform risk deviations give

$$R_{\theta}(\hat{q}) \leq \hat{R}(\hat{q}) + r \leq \hat{R}(q^*) + r \leq R_{\theta}(q^*) + 2r = R_{\theta}^* + 2r.$$

Both conclusions hold on the same event $\mathcal{E}$, with no second failure probability. Since the bound for $\mathcal{E}$ holds at every parameter, the guarantee is uniform in the required sense (a separate probability statement for each parameter).

Step 5: Asymptotic and fallback bookkeeping. For fixed $\delta \in (0, 1)$, $r \rightarrow 0$. If $\epsilon_n \sqrt{n} \rightarrow \infty$, eventually $\epsilon_n \geq 4\sqrt{\log(4/\delta)/(2n)}$, so the guarantee applies and the excess-risk bound tends to zero. For smaller tolerances the defined fallback is exactly fair but has no claimed oracle-risk guarantee. Thus the result does not contradict the lower bound. At equality $\epsilon = 4r$ comparator feasibility is non-strict and remains valid. For small n the risk bound may exceed one and is simply uninformative; no practical constant-optimality or general-class guarantee is asserted.

Meaning of the fixed-confidence limit. For each sufficiently large sample size, the argument gives

$$\Pr_{\theta}\{R_{\theta}(\hat{q}_n) - R_{\theta}^* > 2r_{\delta,n}\} \leq \delta, \quad 2r_{\delta,n} \rightarrow 0.$$

The failure level is fixed in this statement. Thus the vanishing additive term alone does not assert convergence in probability, which would require control of the failure probability tending to zero. In addition, the comparator is exactly fair and the learned rule may use positive tolerance, so signed excess can remain negative. These two distinctions specify the precise asymptotic conclusion of the proposition. $\square$

D Proof of the Utility Envelope

Proof. We prove Theorem 4.

We account separately for the global fairness event and central utility event. The proof uses the local probability inequality proved in Theorem 3 and the Bernoulli concentration calculation established within Proposition 5. For completeness, the concentration argument and the precise event needed from it are given first; no condition $\epsilon \geq 4r$ from the LP theorem is used.

Step 1: confidence bounds for the observed frequencies. For a Bernoulli variable Z of mean p , let $g(t) = \log \mathbb{E} \exp(t(Z - p))$. Direct differentiation gives $g(0) = g'(0) = 0$ and $g''(t) = p_t(1 - p_t) \leq 1/4$, where $p_t = pe^t/(1 - p + pe^t)$. Twice integrating, for either sign of t , yields $g(t) \leq t^2/8$. Independence of the Bernoulli observations therefore implies $\Pr(\bar{Z} - p \geq x) \leq \inf_{t>0} \exp(-tnx + nt^2/8) = \exp(-2nx^2)$, on choosing $t = 4x$. The negative tail follows by applying the same calculation with $t < 0$. For $a \in (0, 1)$, set $r_a = k_a/\sqrt{n}$. A union bound over both groups and both signs gives

$$E_a = \{|\hat{u} - u| \leq r_a, |\hat{v} - v| \leq r_a\}, \quad \Pr_{\theta}(E_a) \geq 1 - a$$

for every θ in the parameter square. The identity $4 \exp(-2nr_a^2) = a$ fixes the confidence constant. Writing $d = u - v$ and $\hat{d} = \hat{u} - \hat{v}$, the triangle inequality gives $|d - \hat{d}| \leq 2r_a$ on this event.

Step 2: a globally fair interpolating predictor. Define

$$s = \min \left\{ 1, \frac{\epsilon}{|\hat{d}| + 2r_{\delta}} \right\}, \quad q^{\text{sh}} = \left(\frac{1-s}{2}, \frac{1+s}{2}, \frac{1-s}{2}, \frac{1+s}{2} \right).$$

Since $r_{\delta} > 0$, the denominator is strictly positive even at $\hat{d} = 0$ or $\epsilon = 0$. Thus $s \in [0, 1]$; all four coordinates belong to the required cube, and this formula is a measurable, parameter-independent learner on the finite training experiment. Its two slopes are both s , its intercept difference b is zero, and consequently its population disparity is $s|d|$. Moreover the error probability at either feature value is exactly $(1-s)/2$: for $X = 0$ the false-positive probability is the lower coordinate, and for $X = 1$ the false-negative probability is one minus the upper coordinate. The known deterministic label relation $Y = X$ in the binary experiment therefore gives

$$\Delta_{\theta}(q^{\text{sh}}) = s|d|, \quad R_{\theta}(q^{\text{sh}}) = \frac{1-s}{2}.$$

On E_{δ} , $|d| \leq |\hat{d}| + 2r_{\delta}$. By the definition of the minimum, $s(|\hat{d}| + 2r_{\delta}) \leq \epsilon$, including the branch $s = 1$. It follows that $\Delta_{\theta}(q^{\text{sh}}) \leq \epsilon$ with probability at least $1 - \delta$ for each θ . This verifies the full uniform fairness requirement, not merely fairness at the central law.

Step 3: the attainable central utility. At $\theta = 0$, the separate event E_{η} implies $|\hat{d}| \leq 2r_{\eta}$. Hence on an event of probability at least $1 - \eta$,

$$s \geq \min \left\{ 1, \frac{\epsilon}{2(r_{\delta} + r_{\eta})} \right\}, \quad \frac{1}{2} - R_0(q^{\text{sh}}) \geq \min \left\{ \frac{1}{2}, \frac{\epsilon}{4(r_{\delta} + r_{\eta})} \right\}.$$

The last display and Step 2 prove that this improvement belongs to the set whose supremum defines G_n . Substituting $r_a = k_a/\sqrt{n}$ gives the stated lower bound. The two event probabilities serve distinct requirements; their intersection is not asserted to have probability $1 - \min\{\delta, \eta\}$.

Step 4: a converse for every admissible learner. Let $\gamma > 0$ be any element of the defining set, and fix a learner attaining its required probability. Set $c = \sqrt{8/3}$, $a = 1 - \eta - \delta > 0$, and

$$h = \frac{a}{2c\sqrt{n}}.$$

Since $a < 1$ and $n \geq 2$, $h < 1/(2\sqrt{16/3}) = \sqrt{3}/8 < 1/4$; this is an admissible radius in Theorem 3. If $\epsilon \geq 2\gamma h$, direct rearrangement gives $\gamma \leq \epsilon/(2h) = c\epsilon\sqrt{n}/a$. This is at most $2c(1 - \eta)\epsilon\sqrt{n}/a^2$, because $a < 1 - \eta < 2(1 - \eta)$.

In the remaining branch $\epsilon < 2\gamma h$, Theorem 3 applies with its strictly positive denominator. Using both the learner's utility probability and the upper bound, even if the displayed bound exceeds one, gives

$$(1 - \eta) \left(1 - \frac{\epsilon}{2\gamma h} \right) \leq \delta + c\sqrt{n}h = \delta + \frac{a}{2}.$$

Since $(1 - \eta) - \delta = a$, subtracting the right-hand terms leaves

$$\frac{a}{2} \leq \frac{(1 - \eta)\epsilon}{2\gamma h}.$$

All divided quantities are positive; multiplying by $2\gamma h/a$ now gives

$$\gamma \leq \frac{(1 - \eta)\epsilon}{ah} = \frac{2c(1 - \eta)}{a^2} \epsilon\sqrt{n}.$$

Thus both branches yield the same upper bound on every positive attainable improvement. Improvement zero also satisfies it.

Step 5: supremum and boundary bookkeeping. The defining set is nonempty because the constant predictor has central risk $1/2$ and is uniformly exactly fair. Every member is at most $1/2$ by definition (and no classifier can improve on zero risk). The upper bound applies memberwise without assuming that an optimal learner or a maximizing improvement exists. Taking the supremum establishes its claimed minimum with $1/2$. At $\epsilon = 0$, Step 2 gives $s = 0$, while Step 4 excludes every positive attainable improvement; thus $G_n = 0$ exactly. For arbitrary large tolerance the same formulas remain valid and the lower bound can reach $1/2$. All constants are finite at each fixed permitted confidence pair; their divergence as $\delta + \eta$ approaches one is retained rather than hidden. Finally the lower expression is bounded below by a positive confidence-dependent constant times $\min\{1, \epsilon\sqrt{n}\}$, and the upper expression is bounded above by another such constant times that quantity. This is the asserted order statement, with no claim of optimal constants or optimal confidence dependence. $\square$

E Proof of the Multigroup Characterization

Proof. We prove Theorem 2.

Only the general polynomial-constraint part of Theorem 1 is used. The binary two-group example in that result is not a premise. We verify its general hypotheses from the multigroup model assumptions explicitly before invoking it.

Step 1: the universal coefficient equations. For group $a \geq 2$, its signed prediction-rate difference from group 1 is

$$F_a(\theta, q) = p^\top(q_a - q_1) + \theta_a^\top B^\top q_a - \theta_1^\top B^\top q_1.$$

If q is fair for all parameters, evaluating at the central parameter gives $p^\top q_a = p^\top q_1$. Fix a coordinate ℓ of θ_a and vary it over its open interval, holding every other coordinate at zero. The resulting affine scalar function is identically zero, so its coefficient $(B^\top q_a)_\ell$ vanishes. Repeating for all coordinates and all $a \geq 2$ gives $B^\top q_a = 0$ in those groups. Varying a coordinate of θ_1 in the equation for group 2 gives $B^\top q_1 = 0$. The availability of group 2 uses $g \geq 2$. Conversely these coefficient equations and common base means set every displayed difference to zero for every θ . This proves the claimed equality of the universal kernel with K_B . It uses independent variation of group coordinates on an open product domain, rather than a single linked path of marginals.

Step 2: the dimension inside the prediction cube. Let $V = \ker B^\top$. Full column rank in the multigroup model gives $\dim V = m - r$. Zero column sums imply $\mathbf{1} \in V$, and the linear functional $\ell(q) = p^\top q$ satisfies $\ell(\mathbf{1}) = 1$. In particular ℓ is not identically zero on V . On V^g , consider the linear map

$$T(q_1, \dots, q_g) = (\ell(q_2) - \ell(q_1), \dots, \ell(q_g) - \ell(q_1)) \in \mathbb{R}^{g-1}.$$

It is onto: for any vector $z \in \mathbb{R}^{g-1}$, choose $q_1 = 0$ and $q_a = z_{a-1} \mathbf{1}$ for $a \geq 2$, temporarily without cube restrictions. Then $T(q) = z$. Rank-nullity therefore gives $\dim \ker T = g(m - r) - (g - 1) = g(m - r - 1) + 1$. The all-coordinate value $q^c = 1/2$ belongs to this linear subspace and to the ordinary interior of the cube. Every sufficiently small vector in $\ker T$, added to q^c , remains in the cube. Thus the intersection K_B contains a relatively open neighborhood in $\ker T$, so its affine hull is $\ker T$. This justifies the dimension assertion for the constrained set, not only the unconstrained equations.

Step 3: reduction to the general statistical kernel theorem. The parameter space $(-\rho, \rho)^{gr}$ is nonempty, open and connected. Each group has a fixed finite number n_a of observations in a finite alphabet. Its ordered sample space, and the product sample space across groups, are finite. By independence the probability of any complete sample is a product of factors $p_j + (B\theta_a)_j$, hence a polynomial and therefore continuous function of the parameter. Positivity in the multigroup model and $\mathbf{1}^\top B = 0$ ensure these are valid probability vectors of total mass one throughout that space. The predictor cube is nonempty and compact, and the learner is a fixed Borel Markov kernel by the stipulated learner model. The differences F_a form a finite matrix of affine, thus polynomial, functions multiplying the coordinates of q . All hypotheses of the general part of Theorem 1 are now verified. Its probability characterization applies with the universal kernel identified in Step1 and yields the exact fairness equivalence in the statement, including arbitrary learner randomization. No restriction to deterministic outputs has been introduced.

Step 4: the central label-approximation problem. For a known binary label y_j , the prediction error conditional on feature j is q_j when $y_j = 0$ and $1 - q_j$ when $y_j = 1$. For $q_j \in [0, 1]$ this is precisely $|q_j - y_j|$. At the central law every group therefore has the same single-vector risk functional $C(q) = \sum_j p_j |q_j - y_j|$. The feasible set $[0, 1]^m \cap V$ is nonempty because it contains all constant vectors, and is compact because it is closed in a compact cube. The continuous functional C consequently attains its minimum $L = L(p, B, y)$ at some fixed vector q^* . Every group vector of a point in K_B is separately feasible for this minimization, so $C(q_a) \geq L$. Since the test group weights are positive and sum to one, $R_0(q) = \sum_a w_a C(q_a) \geq L$ on K_B . Conversely the symmetric tuple $(q^*, \dots, q^*)$ lies in K_B : its base means agree and every vector is annihilated by $B^\top$. Its central risk equals L . Thus the minimum central risk over the full multigroup kernel equals this single-vector minimum. For interpretation, positivity of every p_j implies $C(q) = 0$ exactly when $q = y$. Hence this minimum is zero precisely when $B^\top y = 0$; label information orthogonal to the uncertainty directions can survive exact fairness.

Step 5: exact risk cost over all uniformly fair learners. For any learner satisfying the uniform exact-fairness requirement, Step3 gives $\Pr_0(q \in K_B) \geq 1 - \delta$. The deterministic inequality on K_B proved in Step4 yields the stated high-probability risk floor. Because risk is nonnegative even outside K_B ,

$$\mathbb{E}_0 R_0(q) = \mathbb{E}_0 [R_0(q) \mathbf{1}_{K_B}] + \mathbb{E}_0 [R_0(q) \mathbf{1}_{K_B^c}] \geq L \Pr(K_B) \geq (1 - \delta)L.$$

To prove equality of the infimum, construct a learner independent of the observations. With probability $1 - \delta$, output $(q^*, \dots, q^*)$; with probability δ , output $(y, \dots, y)$. All parameters p, B, y specifying this mixture are known under the stated known-geometry assumptions, so the learner is parameter-independent and admissible. The first output is fair for every parameter; therefore the learner has exact-fairness probability at least $1 - \delta$ everywhere, regardless of whether its second output is fair there. At the center the second output predicts the known labels perfectly and has risk zero. The expected central risk of the mixture is exactly $(1 - \delta)L$, which matches the lower bound. This is an attainable infimum, and does not require estimating a population parameter from the observations.

Step 6: a nonconstant kernel with varying cost. In the specified four-category family, B has rank one for every $\lambda \in [0, 1]$, its column sum is zero, and all conditional probabilities $1/4 + \theta_a B_j$ are positive when

$|\theta_a| < 1/4$. For a single group put $d_1 = q_2 - q_1$, $d_2 = q_4 - q_3$. The constraint $B^\top q = 0$ is $d_1 + \lambda d_2 = 0$, whereas

$$C(q) = \frac{q_1 + (1 - q_2) + q_3 + (1 - q_4)}{4} = \frac{1}{2} - \frac{d_1 + d_2}{4} = \frac{1}{2} - \frac{(1 - \lambda)d_2}{4}.$$

Every cube vector has $d_2 \leq 1$, and $1 - \lambda \geq 0$. It follows that $C(q) \geq (1 + \lambda)/4$ on the feasible set. The vector

$$q^* = \left(\frac{1 + \lambda}{2}, \frac{1 - \lambda}{2}, 0, 1 \right)$$

has all coordinates in the cube, satisfies $d_1 = -\lambda, d_2 = 1$, and attains that lower bound. Choosing it in each group also satisfies the shared-mean constraint; in fact its base mean is $1/2$. At the center the perfect label vector y , used identically in all groups, has prediction mean $1/2$ in every group and zero risk, so the population fair oracle has risk zero as asserted.

Step 7: boundaries and conclusion. At $\lambda = 0$ the cost equals $1/4$; at $\lambda = 1$ it equals $1/2$. The exhibited optimizer is nonconstant in both cases, and the kernel dimension is the same $2g + 1$ throughout. Thus dimension alone does not determine retained utility. All confidence claims include $\delta = 0$, when the attaining mixture reduces to the fixed kernel optimizer. No division by δ or group weights occurs. The cube-relative dimension argument also covers maximal rank $r = m - 1$, when $\dim V = 1$ and the kernel dimension is one. Combining the coefficient characterization, its statistical probability consequence, the matching risk construction, and the explicit family proves every part of the theorem under the stated known-geometry assumptions. $\square$

F Experimental Details

F.1 Population calculations and sample generation

All simulations use NumPy 1.26.4 and SciPy 1.10.1 on CPU. Linear programs use the HiGHS solver with box constraints. The scripts evaluate risk and disparity from the stipulated population probabilities, rather than estimating them from a held-out sample. For the binary studies, independent binomial counts generate the two empirical frequencies. Every trial uses a fresh NumPy `SeedSequence`; its integer components include the study identifier, each grid index, and the trial index. This construction fixes reproducibility without sharing a random stream accidentally between cells.

The utility study uses seed components $(314159, i_n, i_t, i_\theta, i_{\text{trial}})$, 500 trials per cell, and the population pairs $(0, 0)$, $(0.15, -0.15)$, $(0.4, 0.4)$, $(0.4, -0.4)$. For every rule the program records gain, risk, population disparity, and the fairness event. The shrinkage risk identity $R_\theta = (1 - s)/2$ is evaluated directly, as is the exact disparity $s|u - v|$. Constants and the label rule supply population baselines. The lower tenth percentile is computed using the inverse empirical distribution convention. Its error bars use 400 seeded percentile-bootstrap replicates. These intervals describe finite Monte Carlo variability; coverage at a distributional atom is not guaranteed.

The oracle study uses seed components $(271828, i_n, i_\theta, i_{\text{trial}})$ and 200 trials per cell. The four-coordinate objective is the linear form in (2); the empirical program has either the tightened band $\epsilon - 2r_\delta$ or band zero. A separate program with true probabilities and band zero supplies R_θ^* . The recorded joint event checks both $\Delta_\theta(\hat{q}) \leq \epsilon$ and $R_\theta(\hat{q}) - R_\theta^* \leq 2r_\delta$, with numerical comparison tolerance 10^{-9} . The plotted mean signed excess is descriptive and does not replace this event check.

F.2 Multigroup programs and robustness perturbation

The kernel study uses seed components $(161803, i_\lambda, i_\theta, i_{\text{trial}})$. Each of three groups supplies an independent multinomial count vector of size 128. The empirical-equality comparator optimizes 12 prediction probabilities with two acceptance-equality constraints and equal test weights. The known-kernel rule copies $((1 + \lambda)/2, (1 - \lambda)/2, 0, 1)$ into every group. The mixture selects the copied perfect label rule with probability 0.1 using an independent draw in the trial’s random stream. Population equality uses a numerical threshold 10^{-8} .

At the center, the mixture’s risk is either the deterministic kernel cost or zero, and both branches satisfy demographic parity. At the off-center parameter, the label rule can fail equality. The study records this distinction rather than assigning every mixture outcome an assumed fairness probability. The empirical-equality program has a different feasible set from the population-uniform kernel problem; its lower observed risk is interpreted together with its population disparity.

The misspecification study uses seed components $(141421, i_n, i_\tau, i_{\text{trial}})$, two sample sizes, five amplitudes, and 200 trials per cell. Its populations are strictly positive for all tested amplitudes. The expanded-kernel control solves

$$\min_{q \in [0,1]^4} p^\top |q - y| \quad \text{subject to } B^\top q = C^\top q = 0.$$

The objective is linear because y is binary. The resulting rule $(1, 0.5, 0, 1)$ is supplied to all groups. The script checks the nominal identity $\Delta = \tau/2$, equality of the expanded and constant controls, and equality of the nominal rule at $\tau = 0$. The maximum pairwise group difference is used as the multigroup disparity. Deterministic control rows are repeated only to align the result records with the empirical comparator; their intervals are not interpreted as uncertainty estimates.

F.3 Uncertainty and reproducibility

For N independent stochastic trials with event frequency $\hat{p}$, the reported pointwise Wilson interval uses $z = 1.9599639845$, center

$$\frac{\hat{p} + z^2/(2N)}{1 + z^2/N}, \quad \text{and half-width} \quad \frac{z\sqrt{\hat{p}(1-\hat{p})/N + z^2/(4N^2)}}{1 + z^2/N}.$$

Mean-risk intervals use the sample standard deviation divided by $\sqrt{N}$, multiplied by the same z . These normal-approximation intervals are summaries of repeated simulations, not finite-sample confidence theorems for the learning procedure. No multiplicity correction is applied across grid cells. The bootstrap for the utility quantile is separate from the Wilson event intervals.

The accompanying experiment files contain every trial, grid configuration, population metric, and summary, together with execution logs and the scripts. Scientific runs were preceded by small pilots. A later figure-only execution copied the original result data without generating additional statistical trials; the figure revisions changed presentation rather than sample size. The utility and oracle figures compare observed behavior with analytic bounds, while the kernel figure separates actual learner risks from the exact cost among uniformly fair procedures. Agreement on a finite grid supports implementation and interpretation but cannot replace the universal proofs.